# Three random decks

Mathematical overview | Draft and unpiloted | 23 October 2026

## The mathematical core

For independent uniform draws from decks of m and n physical cards, each of the mn card pairs is equally likely. A win probability is the number of winning physical pairs divided by mn. Repeated values remain separate cards. Replacement restores the same probabilities in each round.

**A strict cycle exists:** A=(2,4,9), B=(1,6,8), C=(3,5,7) give P(A>B)=P(B>C)=P(C>A)=5/9. Thus pairwise advantage need not be transitive, and deck totals do not determine it. Duplicating every card equally preserves probabilities; duplicating only selected cards need not.

**Small-deck limit:** Three cards per deck is the minimum for a strict three-way majority cycle among equal-size fair decks with no values shared across decks. For two-card decks, writing each in nondecreasing order, one wins more than half the pairs exactly when both its corresponding values are larger. That relation is transitive. One-card comparisons are transitive too.

When 1–9 are used once in three three-card decks, all three cyclic advantages cannot be at least 6/9. A finite winner-only run also cannot establish with certainty which of the original decks has the advantage: every finite win/loss sequence has positive probability under either direction.

## What the activity develops

**K–1:** Compare cards and organize all pairs; contrast pair counts with a six-round game. Fractions are optional.

**Grades 2–3:** Construct cycles, track multiplicity, and reason about what a finite experiment can establish.

**Grades 4–5:** Certify exact probabilities and complete searches, then prove small-deck and strong-cycle obstructions.

## From trials to explanation

The launch demonstrates one draw and its recorded winner, leaving the complete pair collection open. Games provide data; a proposed cycle is a conjecture until all relevant pairs are counted. Order arguments establish impossibility beyond the tested examples.

# Three random decks
# Facilitator guide

**Week 24. Draft and unpiloted. Unscheduled library slot.** This adult guide accompanies final student packets F24-K-v3, F24-23-v2 and F24-45-v3 (8, 7 and 7 pages). It keys all 22 numbered problems. The mathematical topic is nontransitive dice, implemented with visible, equally likely cards.

## Purpose and practical preparation

Children compare all possible pairs, discover a cycle of advantages, and distinguish an exact chance from the outcome of a short game. They then alter decks and explain limits. Three equal-size fair decks can each have an advantage over another without any being strongest against both others.

**For each of three tables:** three labeled opaque bags or separately shuffled, identical-backed decks; three cards in each deck; nine spare cards; eighteen counters in two distinguishable styles, which record K-1 Problem 4's rounds (one counter per round in the style of the round's winner, six rounds against each of A, B and C); blank paper and pencils. Set A=(2,4,9), B=(1,6,8), C=(3,5,7). Across the room this is nine bags, 27 original cards, 27 spare cards and 54 counters. Provide large numeral/dot cards if useful, and do not depend on color to distinguish decks.

**Making the cards.** Cut index cards in half and use small paper lunch bags. Write each value large, with its dots for the K-1 table as on the student pages: originals in marker, spares in soft pencil so they can be erased and rewritten. Cards cut from one stock are indistinguishable by touch. The room needs 58 cards: 27 originals, 27 spares and the cards 0, 10, 11 and 12, in marker.

**Extra preparation by route:** six-card comparisons need two copies of every original card, eighteen physical cards per table. Use the nine spares as the second copy. For four-card comparisons, keep one extra copy of each smallest value. The middle Problem 6 pool needs 1-12, each once; add cards 10,11,12. Middle Problem 4 offers 0 for A's empty card; add card 0. Upper Problem 3 may repeat a value three times, as in B=(3,3,3): erase the nine spares and write its decks on them. The student illustrations are recording prompts, not cutting templates.

**Once, about 30 minutes:** write the 58 cards and label the bags; keep them for return visits. **Before each session, 10-15 minutes:** print selected Letter pages single-sided at 100%; check every bag has its intended cards; practice returning the cards after a draw. Use the adult matrices on page 3 to verify labels, then set them aside. Cards must be indistinguishable by touch. Mix decks separately. No looking, paired shuffles, rerolls or discarding inconvenient outcomes.

## Shared launch in about five minutes

Allow a few minutes to handle and compare cards. Show one card from each of two bags, say which larger value wins, return both, then mix separately. Next lay out one definite pair and record both values and its winner. Ask: "How could we find every possible pair, including ones we might never draw today?" Begin the group tasks without revealing all nine pairs or the cycle.

**Model limits.** Each physical card has equal chance and draws from different bags are independent. Replacement restores the same chances each round. A fair six-sided die can represent each three-card deck by repeating each value twice; sticker-modified dice are not automatically fair. No real-money stakes are needed.

## Table time and readiness gates

| Time | Plan to adapt |
|---|---|
| Launch, about 5 min | After the opening run-around: handle cards, then the shared comparison and replacement launch. |
| Tables, first 15-20 min | Build complete comparisons (K-1 Problems 1-3, Grades 2-3 Problems 1-2, Grades 4-5 Problem 1) with pooled records, or start a first construction. |
| Tables, next 15-20 min | Choose construction, changed-card or impossibility work. Optional reset: one child points to each pair while another records. |
| Sharing, last few min | Share the cycle and one reason a short game can disagree; note the next question; restore decks. |

**K-1 core:** Problems 1-3 together build one shared cycle investigation: split card-pair cases among children, then pool a checkable complete nine-pair record for each comparison. Each child need not record all 27 outcomes. Move on when the cycle is clear rather than stretching each page artificially. Problem 4 tests the meaning of advantage; Problems 5-6 are construction reserves; Problems 7-8 are optional impossibility tasks.

**Grades 2-3 core:** In Problem 1, share the three deck comparisons among children or pairs and pool a complete, checkable nine-pair record for each comparison. No child needs 27 individual records. Problem 2 uses short runs of draws to distinguish chance from certainty: an adult secretly puts two of the decks in two bags, a child draws from each, and the adult announces only which bag won. It needs no fresh exhaustive tally. For a new arrangement in Problem 3, share the comparison work again and check the pooled cycle. Choose Problems 3-4 for modifying decks, Problem 5 for physical duplicates, or Problem 6 for a harder exact-total construction. Problem 7 can close the session with a short obstruction.

**Grades 4-5 core:** Problem 1, then either the complete-search route in Problem 2, the duplication route in Problems 3-4, or the two-card proof route in Problems 5-6. The last problem is a substantial reserve. One complete search can use most of the table time.

## Access without reducing the mathematics

**Concrete gate:** compare two quantities up to nine and keep track of which decks supplied them. An adult reads and writes; children may pair physical cards, point to dot quantities, or place one counter for each pair. Keep cards in stable rows or use tactile separators. Do not require fraction notation.

**Counting gate:** distinguish a card pair from a value or a single winning card. Nine pairs come from three choices against each of three choices. A child can show all nine in objects or drawings before writing 3 x 3. Offer a row-by-row organization only if the child needs a way to avoid omissions.

**Probability gate:** "wins more pairs" describes all equally likely outcomes, not what must happen next. With duplicate values, mark physical cards 2a,2b on the adult record without making one likelier. For older children use favorable pairs divided by total pairs, with denominators 9,36,16 or 4 as appropriate.

**Proof gate:** distinguish "our attempts failed" from a reason every attempt fails. For two-card problems, physically compare the deck holding the smallest card. For the general rule, align each deck's smaller and larger cards. For exhaustive searches, use an explicit case list. Let children stop with a demonstrated, understood result.

**Held hints and sampling.** Ask what has been tried, then offer one hint and wait. Never keep drawing until results match the prediction and call that proof. A losing run does not invalidate a correct nine-pair count; an unexpectedly successful run does not certify it.

# K-1 Problems 1 to 3 and the shared comparison key

Each cell below names the winning deck for the pair given by its row and column. These adult answer diagrams account for every pair once. Pool the children's assigned cases into one complete record per comparison. Keep this key hidden until needed; the shared record can use objects, counters or adult scribing.

## Problem 1 A against B

| A card / B card | 1 | 6 | 8 |
|---|---|---|---|
| 2 | A | B | B |
| 4 | A | B | B |
| 9 | A | A | A |

**Solution:** A wins (2,1),(4,1),(9,1),(9,6),(9,8): five pairs; B wins four. **Hints:** first look at the row of pictured pairs that share one A card; later ask whether every pictured pair has a circled winner. **Gate:** compare quantities and record both members of a pair. A physical nine-pair demonstration is sufficient.

## Problem 2 B against C

| B card / C card | 3 | 5 | 7 |
|---|---|---|---|
| 1 | C | C | C |
| 6 | B | B | C |
| 8 | B | B | B |

**Solution:** B wins (6,3),(6,5),(8,3),(8,5),(8,7): five pairs; C wins four. **Hints:** first compare the role of B's smallest and largest cards; later check three pairs for each card. **Gate:** retain complete-pair coverage from Problem 1. No fraction writing is needed.

## Problem 3 C against A

| C card / A card | 2 | 4 | 9 |
|---|---|---|---|
| 3 | C | A | A |
| 5 | C | C | A |
| 7 | C | C | A |

**Solution:** C wins (3,2),(5,2),(5,4),(7,2),(7,4): five pairs; A wins four. Thus A beats B, B beats C, C beats A. None beats both others. Here "beats" always means wins more possible pairs, not every draw. **Hints:** first mark an arrow from each favored deck to the other; later ask which arrow enters each deck. **Gate:** hold all three comparisons in view. Stop once the cycle is understood.

**Why these counts give exact chances:** three equally likely cards independently paired with three others give nine equally likely outcomes. Each favorable direction has chance 5/9, and the reverse has 4/9. None of the original decks shares a value with another. These same matrices key both older bands' Problem 1.

# K-1 Problems 4 to 6

### Problem 4 Choose an advantage and play

**Solution.** Against A choose C; against B choose A; against C choose B. No chosen deck must win more of its six rounds. For example C loses to A whenever C draws 3 and A draws 9, and replacement allows that pair to recur all six times. More generally the favored deck can lose all six with chance $(4/9)^6$, which is positive. A 3-3 tie is also possible. Keep the exact pair count and the observed six-round record separate.

**Hold these hints.** First: "Could this losing pair happen again after we put it back?" Later: lay out one legal sequence of six losses.

**Readiness and stopping point.** Children need the replacement rule, not powers or fractions. The partner holds A, then B, then C; the eighteen counters record the rounds, one per round in the winner's style. Every legal six-round outcome is accepted. Do not make them retry until the favored deck comes out ahead.

### Problem 5 Fill the third A card

**Solution.** Only 9 works. A's fixed 2 and 4 each beat only B's 1, giving two wins. A third card 3,5,7,9 contributes respectively 1,1,2,3 wins. Thus A totals 3,3,4,5 out of nine; more than B requires at least five.

**Hold these hints.** First: "What do the unchanged cards already win?" Later: compare just the candidate against B's three cards.

**Readiness and stopping point.** Count all four candidates; a card 7 still leaves A behind, 4 wins to 5. A complete list is a sufficient explanation.

### Problem 6 Every single swap

**Solution.** The table lists B's new wins out of nine after one swap from the original decks. Reset after each trial; swaps are not cumulative.

| Original A card to swap | Swap with B 1 | Swap with B 6 | Swap with B 8 |
|---|---|---|---|
| 2 | 5 works | 2 | 1 |
| 4 | 6 works | 3 | 2 |
| 9 | 9 works | 6 works | 5 works |

Exactly five swaps work: A2/B1, A4/B1, A9/B1, A9/B6 and A9/B8. The nine cells exhaust three choices from A times three choices from B. In the 9-for-1 swap the new A=(1,2,4) and B=(6,8,9), so B wins every pair.

**Hold hints.** First: label which original cards were exchanged. Later: try all three partners for a fixed A card before changing it. **Gate:** reset to the pictured decks every time. Allow adult recording. Finding all five and explaining why all nine swaps were considered completes the problem.

# K-1 Problems 7 and 8

### Problem 7 Can the two decks split their wins equally

**Solution.** Impossible. Any two three-card decks made from 1-6 have nine possible pairs, and every pair has exactly one winner because the six values are all different. Equal win counts would have to split nine into two equal whole-number piles, which is impossible. Physically put nine counters in a row and assign one to the winner of each pair; equal piles always leave one counter unpaired. This applies to every way of dividing the cards.

**Hold these hints.** First: "How many winners are there in total, counting every pair once?" Later: try to make two equal piles from those nine counters.

**Readiness and stopping point.** Children need the idea of an odd total, not division by two. An oral explanation with paired counters is complete; unsuccessful deck searches alone are not.

### Problem 8 Can three two-card decks make a cycle

**Solution.** Impossible. Find the deck holding card 1. Against either other deck, its 1 loses both pairs. Its other card can win at most two pairs. Therefore this deck can never win more than half of the four pairs against another deck. A cycle would require every deck, including this one, to have a majority win against somebody. That cannot happen. The identity A, B or C of the deck holding 1 does not matter.

**Hold these hints.** First: "Follow the deck containing 1." Later: lay out the four pairs against one other deck and mark the two unavoidable losses.

**Readiness and stopping point.** This is a concrete proof for the pictured six distinct cards. A drawing of the two inevitable losses and two remaining possibilities is enough. Save the general two-card theorem for older children if it distracts from the concrete obstruction.

| The deck holding 1 | Other deck: first card | Other deck: second card |
|---|---|---|
| Its card 1 | Loses | Loses |
| Its other card | Might win | Might win |

# Evidence worth sharing at the end

Invite a child to show either why no deck is strongest in the original example, why a favored deck can lose a short game, or why a two-card cycle cannot happen. Each is a real mathematical conclusion at a different level. One understood counterexample or impossibility argument is a better stopping point than rushed completion of every page.

**Do not extend the odd-pair proof silently.** It depends on all pairs being decisive. If equal values across decks were allowed, tied outcomes would need their own count. Likewise, the smallest-card argument assumes equally likely physical cards. The printed six-card pool and shared rules meet both conditions.

# Grades 2-3 Problems 1 to 3

### Problem 1 Compare all three decks

**Solution.** A beats B in five pairs to four; B beats C in five to four; C beats A in five to four. No deck beats both others. Page 3 gives every pair and winner. The three sets of nine pairs, not a game tally, certify the answer.

**Hold these hints.** First: "Which opponent does each deck struggle against?" Later: make one complete comparison before drawing the three arrows.

**Readiness and stopping point.** A complete pair record can be physical or written. Fractions are optional; the strict majority of nine is five.

### Problem 2 Does a hidden winning streak establish certainty

**Solution.** No, even six left-bag wins are possible when the left bag has the disadvantage. In every matchup of the original decks, one side has chance 5/9 and the other 4/9 on each fresh draw. With replacement, a left-winning pair can recur as often as desired. Thus any fixed finite streak of n left wins has positive chance $(4/9)^n$ even under disadvantage. More generally any finite sequence containing w left wins and l right wins has positive chance under both assignments: $(5/9)^w(4/9)^l$ in one direction and $(4/9)^w(5/9)^l$ in the other. Winner-only data cannot give logical certainty in finitely many trials.

**Hold these hints.** First: "Could the weaker deck win this one draw?" Later: repeat one such pair after replacing the cards.

**Readiness and stopping point.** Distinguish certainty from evidence: a streak can make an advantage more plausible under a specified model; it does not prove it. Do not say that observations teach nothing. The visible student task reveals only the winner, not card values.

### Problem 3 Two new fixed-maximum cycles

**Solution.** The two arrangements different from Problem 1 are (i) A=(1,5,9), B=(3,4,8), C=(2,6,7), with directed counts (5,5,5); and (ii) A=(2,3,9), B=(1,6,8), C=(4,5,7), with counts (5,5,6). All nine cards occur once and 9,8,7 stay in their assigned decks. Along with the original these are the only three arrangements, ignoring order within a deck. The complete case certificate on page 9 checks that last statement; this problem only asks children to find the two new ones.

**Hold these hints.** First: "Keep a record when two of the three arrows work." Later: compare what each low card beats, since the three large cards stay fixed.

**Readiness and stopping point.** This is a sustained search, not a quick extra. Permit most of the work period here. Do not ask for a third new solution: there are only two. Choosing which card to move remains the child's work.

# Grades 2-3 Problems 4 to 7

### Problem 4 Every replacement that preserves the cycle

**Solution.** In A=(2,x,9), only x=2 or x=4 works. Directed counts below mean A over B, B over C, C over A, all out of nine.

| x | A beats B | B beats C | C beats A | Cycle? |
|---|---|---|---|---|
| 0 | 4 | 5 | 6 | No |
| 2 | 5 | 5 | 6 | Yes |
| 4 | 5 | 5 | 5 | Yes |
| 9 | 7 | 5 | 3 | No |
| 10 | 7 | 5 | 3 | No |

**Why complete:** all five allowed choices are listed. For x=0, A loses its first edge; for 9 or 10, C loses its edge over A. Two cards labeled 2 remain two separate physical cards. **Hints:** first identify unchanged pairings; later count only the changed card's wins. **Gate:** track multiplicity, not just distinct labels.

### Problem 5 Double every original card

**Solution.** Each original pair is replaced by four physical pairs, because each of its two cards has two copies. Thus every favorable comparison becomes 20 wins to 16 out of 36. The favored directions remain A over B, B over C, C over A; 20/36 =5/9. There are 36 equally likely physical pairs, not nine.

**Hold these hints.** First: label identical cards a and b. Later: list the four copy-pairs arising from one original pair.

**Readiness and stopping point.** A multiplication explanation can replace 36 separate marks after the fourfold correspondence is understood.

### Problem 6 Meet the revised totals exactly

**Solution.** A=(3,5,7), B=(2,4,12), C=(1,9,11). Totals are 15,18,21; all nine values differ and lie in 1-12. Leave 6,8,10 unused. A over B: row win counts 1+2+2=5. B over C:1+1+3=5. C over A:0+3+3=6. Therefore all three required advantages hold, even though the deck totals increase in the opposite order to the first two arrows.

**Hold these hints.** First: try a triple with the required total before testing its wins. Later: a very large card can contribute three wins while leaving room for two small cards.

**Readiness and stopping point.** Use separate columns for total and win count. A correct construction is sufficient; uniqueness is not requested.

### Problem 7 The six-card two-card obstruction

**Solution.** No arrangement works. Whichever deck contains 1 loses two of its four pairs against either other deck, so it can win at most two. It therefore has no majority advantage and cannot participate in the required cycle. This is the same complete concrete argument as K-1 Problem 8 on page 5.

**Hold these hints.** First: track the smallest card. Later: show its two unavoidable losses.

**Readiness and stopping point.** The argument must rule out all arrangements, not just the ones tried. Pair drawings are enough; no general algebra is required.

# Grades 4-5 Problems 1 and 2

### Problem 1 Exact chances and equal totals

**Solution.** Every favorable edge has five winning pairs out of nine: A over B, B over C, C over A. Reverse edges have four. Page 3 lists all outcomes. No deck wins more than half against both others. Each total is 15 and each mean is 5, yet these equal totals coexist with a strict cycle. Totals alone cannot settle which deck has a pairwise advantage: they do not specify how the values interleave.

**Hold these hints.** First: "Does holding the 9 make A beat C?" Later: compare the two smaller A cards with all three C cards.

**Readiness and stopping point.** Require exact counts, not just an observed game outcome. Means are optional; total 15 in all decks is sufficient to discuss the prompt.

### Problem 2 All fixed-maximum arrangements

**Solution.** Exactly the three rows below meet A over B, B over C, C over A. Card order within a deck does not create a new arrangement.

| A | B | C | Directed wins /9 |
|---|---|---|---|
| (1,5,9) | (3,4,8) | (2,6,7) | 5,5,5 |
| (2,3,9) | (1,6,8) | (4,5,7) | 5,5,6 |
| (2,4,9) | (1,6,8) | (3,5,7) | 5,5,5 |

**A complete and inspectable search.** Choose A's two low cards from 1-6, then B's two low cards from the four remaining; C gets the last two. This gives 15 x6=90 cases, with no duplicates because each pair is unordered and the decks are labeled. Page 9 gives a compact certificate eliminating all but the three rows.

Let a,b,c be those low pairs and W(x,y) count wins of x's two cards against y's two cards. A's 9 beats all three B cards, while both low A cards lose to 8, so the first advantage is exactly W(a,b) >=2. Similarly B's 8 beats all three C cards, so the second is W(b,c) >=2. C's 7 beats both low A cards but loses to 9; both low C cards also lose to 9. Thus the last advantage is exactly W(c,a) >=3. These three small comparisons exhaust all nine pairs for each edge.

**Hold hints.** First: "What choices determine the whole arrangement?" Later: hold A's low pair fixed and test all six choices of B's low pair. The W formulas and case table are adult-held scaffolds, not a starting procedure to impose.

**Readiness and stopping point.** Complete case coverage is the mathematical aim. A child-made search organized differently is equally valid if it covers every arrangement once or explains its omissions. This task can use the entire exploration period. Accept the three valid examples first, then ask what would make the list complete.

## Complete certificate for fixed maxima

This is the continuation of grades 4-5 Problem 2 and the adult completeness check for grades 2-3 Problem 3. Append 9 to A,8 to B and 7 to C. In every row C's low pair is the two unused cards. The middle column retains exactly W(a,b) >=2 and W(b,c) >=2; the last also requires W(c,a) >=3.

| A low pair | B low pairs passing first two edges | B low pairs passing all three |
|---|---|---|
| (1,2) | None | None |
| (1,3) | None | None |
| (1,4) | None | None |
| (1,5) | (3,4) | (3,4) |
| (1,6) | (2,5), (3,4), (3,5), (4,5) | None |
| (2,3) | (1,6) | (1,6) |
| (2,4) | (1,6) | (1,6) |
| (2,5) | (1,6), (3,4) | None |
| (2,6) | (1,5), (3,4), (3,5), (4,5) | None |
| (3,4) | (1,6), (2,5), (2,6) | None |
| (3,5) | (1,6), (2,4), (2,6) | None |
| (3,6) | (1,5), (2,4), (2,5), (4,5) | None |
| (4,5) | (1,6), (2,3), (2,6), (3,6) | None |
| (4,6) | (1,5), (2,3), (2,5), (3,5) | None |
| (5,6) | (1,4), (2,3), (2,4), (3,4) | None |

**How to verify a table row.** For A low pair(1,5), the remaining cards are 2,3,4,6. The six possible B pairs are(2,3),(2,4),(2,6),(3,4),(3,6),(4,6). Against a low B pair containing 6, A's low cards achieve at most one win, so those three choices fail A over B. B=(2,3) leaves C=(4,6), and B=(2,4) leaves C=(3,6); they give zero and one low B-over-C wins, so fail B over C. B=(3,4) leaves C=(2,6): low counts are 2,2,3, so all edges work.

**Why nothing is missing.** Every unordered pair from 1-6 appears in the first column. Every remaining unordered B pair is among that row's six tests; C is then forced. The exact formulas on page 8 make each elimination equivalent to failing a required edge. The last column therefore leaves exactly the three displayed solutions, and each was directly checked in nine-pair comparisons.

**Use with children.** This table is a certificate to reconstruct or inspect after genuine search. It is not necessary to copy it. If children need a smaller search, assign different A pairs to partners and share their complete B lists. The adult can record without taking over the reasoning.

# Grades 4-5 Problems 3 and 4

### Problem 3 A cycle with repeated values

**Solution.** One valid construction is A=(1,4,4), B=(3,3,3), C=(2,2,5). It uses only numbers from 1 to 6, with no value shared between decks; the wording does not require every number 1-6 to be used. A beats B: the two 4 cards each win three, so 6/9. B beats C: each 3 beats both 2 cards, so 6/9. C beats A: the two 2 cards each beat 1, and 5 beats all three A cards, so 5/9. All are strict majorities.

**Hold these hints.** First: try a deck of three equal cards. Later: make another deck with two cards just above that value and one below.

**Readiness and stopping point.** Count physical duplicates separately. This is a construction task, so one fully checked example completes it. Other valid cycles are welcome.

### Problem 4 Duplication changes weights

**Six-card decks.** They duplicate every original card equally. Each original outcome has four physical versions, so the pairwise chances stay exactly the same. **Four-card decks.** Only each smallest value is copied, changing the probabilities. All six directed chances are in the table.

| Comparison | Original three cards | Six cards | Four cards |
|---|---|---|---|
| A over B | 5/9 | 20/36 =5/9 | 10/16 =5/8 |
| B over A | 4/9 | 16/36 =4/9 | 6/16 =3/8 |
| B over C | 5/9 | 20/36 =5/9 | 7/16 |
| C over B | 4/9 | 16/36 =4/9 | 9/16 |
| C over A | 5/9 | 20/36 =5/9 | 10/16 =5/8 |
| A over C | 4/9 | 16/36 =4/9 | 6/16 =3/8 |

**Exact four-card counts.** A=(2,2,4,9) against B=(1,1,6,8) has row wins 2+2+2+4=10. B against C=(3,3,5,7) has 0+0+3+4=7. C against A has 2+2+3+3=10. Reverse counts are complements to 16 because there are no cross-deck ties. Every pairwise chance changes from the original, and B versus C reverses its advantage. The four-card relation is C over A over B, with C also over B; it no longer cycles.

**Hold hints.** First: label duplicate cards with temporary suffixes to count physical choices. Later: compare each row's win count before and after copying the smallest card. **Gate:** use the correct total of equally likely pairs,36 or 16. Counting distinct printed number-pairs as equally likely gives the wrong answer for the four-card decks (B over C would look like 5/9 instead of 7/16). It works for the six-card decks only because every card is copied equally.

**Source limit.** Fair dice are an abstract model of equally likely physical outcomes. These card constructions do not establish that a homemade weighted or sticker-covered die rolls fairly.

# Grades 4-5 Problems 5 and 6

### Problem 5 A complete two-card rule

| Printed pair | First deck wins | Other deck wins | Majority winner |
|---|---|---|---|
| A=(1,5), B=(2,4) | 2 | 2 | Neither |
| C=(2,6), D=(1,5) | 3 | 1 | C |
| E=(4,6), F=(2,3) | 4 | 0 | E |
| G=(1,3), H=(2,6) | 1 | 3 | H |

**Rule.** Sort the cards in each deck: a1 <= a2 and b1 <= b2. With no value shared between decks, A wins more than half the four pairs exactly when a1 > b1 *and* a2 > b2. Both its smaller and its larger card must be larger than the corresponding B card. Repeated values inside one deck are allowed.

**Necessity.** If a1 < b1, then a1 loses to both B cards, so A has at most two wins. If a2 < b2, then both A cards lose to b2, again at most two wins. Equality across decks is excluded, so a strict majority requires both stated strict inequalities.

**Sufficiency.** If both inequalities hold, A wins a1 over b1, a2 over b1, and a2 over b2. These are three different physical pairs, hence a strict majority. The fourth pair may go either way. This proves both directions, not merely a pattern from the four examples.

**Hold hints.** First: place each deck in increasing order and align small with small. Later: ask what happens to an entire row if A's smaller card is too small, or an entire column if A's larger card is too small. **Gate:** understand four equally likely pairs and distinguish a tie in win counts (2-2) from equal card values, which are forbidden across decks.

### Problem 6 Fewest cards per deck

**One card is impossible.** A over B over C over A would mean a > b > c > a, impossible for ordinary numbers. **Two cards are impossible.** By Problem 5, such a cycle would imply a1 > b1 > c1 > a1 and likewise for the larger cards. Either chain is impossible. A shorter alternative: the deck containing the smallest printed value loses at least two of its four pairs against any other deck and cannot have a strict majority.

**Three cards suffice.** The original A=(2,4,9), B=(1,6,8), C=(3,5,7) has the verified cycle 5/9,5/9,5/9. The exact minimum is therefore three cards per deck. This conclusion is for equal-size fair decks, independent draws and no cross-deck shared values; it does not claim the same threshold for unequal sizes, biased draws or a rule that discards tied results.

**Hold hints.** First: translate an arrow into the two-card rule. Later: follow only the smallest card in each deck around the proposed cycle. **Gate:** the coordinate comparison or global-minimum obstruction must be understood, not just the failed search. A complete three-card construction plus the two lower bounds settles the optimum.

# Grades 4-5 Problem 7

### Can all three advantages reach six wins

**Solution: impossible.** This concerns all labeled partitions of 1-9 into three unordered decks of three, not merely the fixed-maxima family in Problem 2. A direct argument rules them all out.

**1. Put the largest card in view.** Rotate the deck names around the proposed cycle so that A contains 9. Cyclic relabeling preserves the requirements A over B, B over C, C over A. Write A=(a,b,9), where a < b.

**2. Use C's required six wins over A.** No C card beats 9. Each of its three cards can therefore win at most two pairs against A. To reach six, every C card must beat both a and b. In particular, all C cards exceed b.

**3. Use A's required six wins over B.** Card 9 contributes exactly three wins. The two smaller A cards must contribute at least three more. If at most one B card were below b, then a and b could together win at most two pairs, both against that one card. Hence at least two B cards must be below b.

**4. The closing edge fails.** Those two B cards lie below b, and every C card lies above b, so those B cards lose all six of their comparisons with C. Only the remaining B card can win, at most three pairs. Therefore B cannot win six against C. This contradicts the proposed cycle.

| Forced order from the two edges | Consequence for B against C |
|---|---|
| At least two B cards < b < every C card | Those B cards win zero pairs |
| Only the remaining B card can win | At most three wins in total |

**Hold hints.** First: "Which deck contains 9, and which deck must beat it?" Later: "If a deck cannot beat 9 at all, what must it do in its other six pairs?" Give the next step only after a fresh attempt.

**Readiness and stopping point.** This uses an upper bound on possible wins rather than a search. Children need to understand why achieving six of six available wins forces every one of those wins. A physical display with two marked low A cards and three C cards can carry the whole proof. The final contradiction is at most three B wins, not merely fewer than six in one example.

**Independent finite audit.** checks.py separately inspected all C(9,3) x C(6,3) =84 x20 =1,680 labeled partitions. It found 15 cycles with all three edges strictly greater than one half, and none with all three counts at least six. The proof above is the explanation; the audit is an error check. No uniqueness or optimality claim for arbitrary biased dice follows from it.

## Extensions and source limits

**Same chances, new labels.** Apply any strictly increasing relabeling to all card values. Every pairwise comparison stays the same, so the cycle remains. Adding 100 to every card is a simple example. This separates order information from the numerical size of totals.

**Copying extension.** Replace every card in a deck by the same number r of copies, and every card in the other by s copies. Every original pair has r x s physical versions, so the winning fraction stays the same. Unequal copying of different values changes their weights, as the four-card example shows.

**Short-game extension.** For the favored original deck, the chance of finishing ahead after six independent rounds is sum over k=4,5,6 of $C(6,k)(5/9)^k(4/9)^{6-k}$, about 45.3%. A 3-3 tie has chance about 30.1%. Thus a strict one-round advantage need not give a greater-than-half chance of being strictly ahead after an even, short series. This optional calculation explains why the six-round game often disagrees with naive expectations; it is not necessary for K-1.

## Verified sources and the boundary of the claims

**Mathematical reference:** Brian Conrey, James Gabbard, Katie Grant, Andrew Liu and Kent E. Morrison, *Intransitive Dice*, *Mathematics Magazine* 89 (2016), 133-143; author version pp. 1-2, verified 2026-10-03: arxiv.org/pdf/1311.6511. It provides the equal-face model, pairwise dominance and Efron's four-die cycle. Its general definition compares win probabilities, allowing ties; the student model forbids cross-deck equal values, making that equivalent to a majority of all pairs. The paper's asymptotic conjectures are not needed here.

The specific three-card example, replacement and swap counts, fixed-maxima completeness certificate, revised exact-total construction, and elementary impossibility proofs are independently derived and checked for this guide. Do not attribute those exact classroom tasks or their effectiveness to the paper.

**Teaching evidence:** Laura Givental, Maria Nemirovskaya and Ilya Zakharevich, *Math Circle by the Bay*, preface printed pp.viii-x (local PDF pp. 9-11), read directly: themes can support multiple depths; dialogue, manipulatives and independent work matter; pace should respond to the group. The local README.md and plans/week-01-classroom-review.md support a common concrete launch and usable representations. Our routes, minute menu and held hints are proposed applications, not evidence that these decks have been piloted.

## Checks and notes after teaching

checks.py is independently implemented and records exact results in checks.json: all nine swaps, all five middle replacements,90 fixed-maxima assignments,1,680 full partitions, repeated-value two-card comparisons, physical duplicate probabilities and the revised total constraints. It finds exactly one solution to the revised 15,18,21 construction, but children are not asked to prove that uniqueness. The final PDF was freshly rendered and every page visually inspected.

After use, record actual tasks tried, children's pair-coverage methods, whether replacement needed reminders, which hints helped, and whether a lucky game was mistaken for proof. Preserve the difference between a prepared extension, an observed attempt and an understood argument. This remains an unscheduled library slot until deliberately selected.

# Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

### Use and return choices

Keep one adult at each table. For K-1 Problems 1-3, split the pictured pairs among the children and pool a complete shared nine-pair page for each comparison. Then bring the three comparisons together to check the cycle; each child need not make 27 records. On return, K-1 can play the advantage game (P4), change one card or swap (P5-6), then try P7-8 if ready. Grades 2-3 also share and pool comparisons for P1 and new arrangements in P3; P2 uses short draws without another exhaustive tally. On return they can choose altered decks, duplication or exact totals (P3-6). Grades 4-5 can choose one complete search, duplication comparison or two-card obstruction (P2-7) as its own visit.